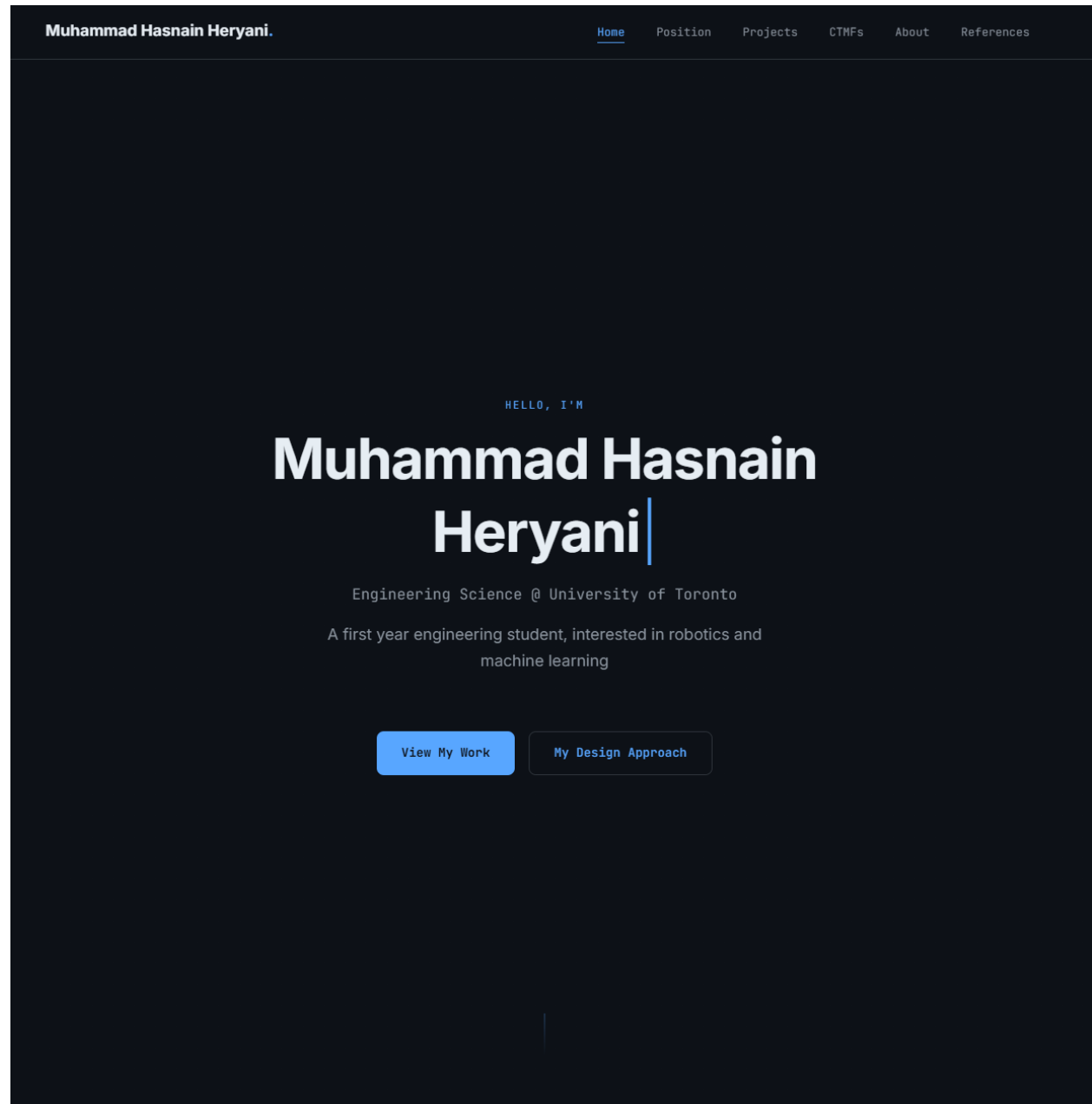


Name: Muhammad Hasnain Heryani

Student Number: 1011884695

UTORID: Heryanim

Website Link: https://hasi-7.github.io/Personal_Portfolio/



POSITION STATEMENT

My Engineering Design Position

A video reflection on my identity as an engineering designer — my values, approach, and how my experiences have shaped my practice.



ANNOTATION

ANNOTATION



My position in engineering design started with how I built the Minecraft 3×3 mechanical door in my original statement, where I broke down a complex system into smaller circuits, used reference designs instead of reinventing everything, tested repeatedly, and adjusted based on observed behavior [4]. That is still my baseline approach, but Praxis II made it more intentional in how I frame and scope real problems [2], [14].

In Praxis II, our team began with a broad opportunity in FPV drone pre-flight diagnostics, then narrowed the scope to motor fault detection after comparing technical feasibility and workflow constraints [10], [14]. That experience clarified a major part of my practice: I value solving a credible problem end-to-end more than proposing an over-ambitious solution that cannot be built or verified. The most important shift for me was not technical complexity by itself, but disciplined scope control tied to stakeholder reality. Meaning, understanding what I am capable of and not over promising something that I cannot confidently deliver while still addressing the stakeholder's problem at hand [2].

Praxis II clarified that my understanding improves most when I verify each core function to its real implemented limits before full integration. For current-analysis testing, we split the concept into sensing and analysis. We first assumed analysis worked and stress-tested sensing in simulation by tuning noise toward realistic conditions. Then we assumed sensing worked and tested analysis using a representative current signal matching sensor output, so the software could be validated independently. That staged verification gave me confidence to move into integration [15]. I do not like integrating when component behavior is still uncertain, because once everything is coupled, failures become harder to isolate and debug.

My position is still evolving, but right now it is this: breakdown problem complexity, test assumptions early, and let evidence drive each next decision [2], [4].

DESIGN PROJECTS

My Work

A collection of engineering design projects spanning human-centred design, structural engineering, and systems design.

STRUCTURAL ENGINEERING

CIV102 — Structural Bridge Design

Designed and built a model bridge to meet specific load requirements, applying structural analysis principles and material science fundamentals.

[View Project](#) →

ACCESSIBILITY DEVICE DESIGN

Roller Support Design to Prevent Smudging for Left-Handed Writing in the EngSci Common Room

In ESC101 Praxis I, our team designed and compared four concepts to help left-handed EngSci students use common-room whiteboards without smudging fresh ink or forcing painful wrist postures. We recommended a roller-support wrist cuff after testing writing speed, visible smudging, ergonomics, and weight.

[View Project](#) →

SYSTEMS DESIGN

Motor Fault Detection via Sound Analysis for Professional FPV Drone Operations

In ESC102 Praxis II, our team designed and prototyped a pre-flight motor diagnostic tool for professional FPV drone operations, replacing subjective audio checks with FFT-based analysis and actionable pass/warn/fail outputs.

[View Project](#) →

DESIGN PROJECT

Roller Support Design to Prevent Smudging for Left-Handed Writing in the EngSci Common Room

ACCESSIBILITY DEVICE DESIGN

CONTEXT

In ESC101 Praxis I, our team addressed the accessibility problem of left-handed whiteboard use in the EngSci common room. After rescoping the problem around fresh marker-ink contact and comparing four concepts through testing and convergence tools, we recommended a roller-support wrist cuff that preserved writing speed while reducing visible smudging and wrist strain.

DETAILS

TYPE Accessibility Device Design
DATE 2025-09 - 2025-12
CTMFS 3 documented

TEAM CREDITS

- Collaborative design work completed with my ESC101 Praxis I project team.
- Teammates: Grant Llang, Melissa Zhang, and Stephanie Zhang.
- Final concept selection and prototype decisions were developed through team consensus.

PROCESS ANNOTATIONS

Summary

This Praxis I project addressed a specific accessibility problem in the University of Toronto's Engineering Science common room with regards to left-handed students. They often dragged their hands through fresh whiteboard ink while writing left-to-right, smudging the ink on the board, transferring marker to their skin, and pushing them into awkward wrist postures. The project was built on stakeholder interviews, narrative accounts, and observation [5], [6]. One left-handed interviewee described propping their hand up with their pinky until it became uncomfortable, while another said whiteboard writing felt "super uncomfortable" because they had to contort their body and wrist to avoid erasing what they had just written [6]. Thus, we found that this was a major ergonomic and communication challenge, where any solution must be ergonomic for the user and not smudge fresh whiteboard ink for the reader.

The primary stakeholders were left-handed EngSci students writing left to right scripts (primarily English) and right-handed students writing right to left scripts, such as Arabic or Hebrew. Secondary and tertiary stakeholders included audiences who needed legible board work and UofT facilities staff who needed the whiteboards left undamaged and free of permanent residue. Our team worked within the strictly defined environment of the Engineering Science common-room whiteboards, treating them as fixed infrastructure, with Expo dry-erase markers assumed as the baseline tool, and ensuring any solution did not add excessive weight, noise, or maintenance burden.

The project was framed as a first-year engineering design project in which we defined measurable requirements [1], [5]. By the final report, with the most important requirements including maintaining baseline writing speed, keeping smudging below a perceivable color difference threshold, keeping wrist flexion or extension within 15 degrees, and keeping total tool weight below 400 g [6].

Outcomes

The final recommendation was the roller-support concept: a wrist cuff and arm platform connected to a polystyrene sphere through a ball transfer unit (Figures 1 and 2). This concept was selected because it balanced smudging reduction, writing speed, ergonomics, and mass better than the alternatives. Three key design decisions supported this outcome: using a polystyrene support ball to reduce ink pickup, using a symmetric cuff to support ambidextrous use, and using a velcro strap to accommodate different wrist sizes without unnecessary added mass.

Our team compared the roller support against three alternatives: a dummy marker (Figure 3), an air-cushion concept (Figure 4), and a marker-mounted platform (Figure 5). Pairwise comparison and Pugh-chart convergence showed that the roller performed most consistently across the full criteria set, rather than maximizing only one criterion [6], [7].

In testing, the roller support and dummy marker both performed well on smudging and ergonomics, while the platform performed worse ergonomically. The air-cushion concept was eliminated due to weight and feasibility constraints. The key difference between the roller support and dummy marker was writing speed: the roller showed a smaller deviation from control speed. This made the roller the strongest recommendation for common-room use, improving on regular whiteboard writing conditions.



Figure 1: A ball transfer unit has many low-friction small balls surrounding the main ball. This lets the main ball spin freely while attached to the arm support platform [6]. Credit: Grant Liang

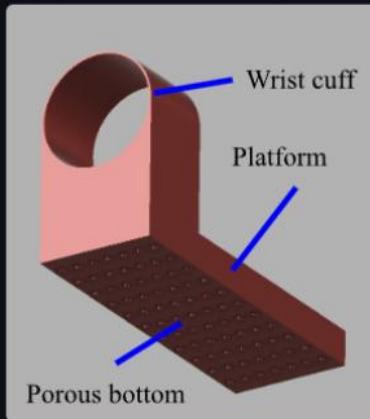


Figure 4: CAD model of air cushion. The platform contains a fan that pushes air through the porous bottom to prevent contact with the whiteboard [7].

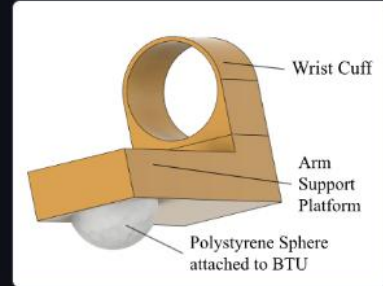


Figure 2: Roller support design. The arm rests on the flat platform and the wrist goes through the wrist cuff. The sphere is attached to the ball transfer unit (BTU) as shown in Figure 4 [6].



Figure 3: Prototype of dummy marker on a whiteboard. The marker that the user holds is taped on the bottom so that it cannot write, and the marker to the left is able to write [7]. Credit: Melissa Zhang



Figure 5: Platform design prototype; the user's hand rests on the attached platform while writing. The platform would be see-through [7]. Credit: Grant Liang

Key outcomes included:

- Converged on the roller-support concept as the best overall balance across smudging, ergonomics, writing speed, and mass [6]
- Eliminated weaker concepts through proxy testing, Pairwise comparison, and Pugh charts [6], [7]
- Demonstrated that the roller preserved writing speed better than the dummy marker while still reducing smudging and discomfort [6]

Process Annotations

What stands out most to me in this project is how much our team improved through the process, especially by Alpha. I am most proud of our Alpha release because we had four real prototype directions with actual reasoning behind each one [7]. A big moment for me was the night before Alpha when we still needed a fourth concept. I went back to the roller-ball idea from our brainstorming notes and realized it had real merit. We had just discussed friction in physics, so I connected that to the idea: if the ball rolls without slipping, it should not drag and displace marker ink the same way. That turned the roller concept from a rough brainstorm into something physically justifiable and testable [6], [7].

Another major takeaway for me was how important proper brainstorming is. At first, I did not expect some of the lecture tools to be that useful, but methods like wishing and our adapted 6-3-5 process directly led to some of our best concepts, including dummy marker and roller support [7]. This changed how I approach ideation now. Some ideas sound unrealistic at first, but if you dig into the mechanism, there can be a strong engineering idea underneath.

The most frustrating part for me was convergence. We had to run many proxy tests to justify eliminating concepts, and because we did not have enough secondary research to eliminate options early, we had to build and test more ourselves. It was time-consuming, but it made the final decision stronger [6], [7]. We initially preferred dummy marker because it was lighter and simpler, but writing-speed testing showed it slowed users down because it still forced a different writing style. The roller support was slightly heavier, but it let users write more naturally while still performing well on key criteria, which is why it became our final recommendation [6].

Overall, this project changed how I think about engineering design. Framing is critical: no matter what problem you are given, you still need to reframe it so you are solving the right thing [5], [6]. Also, good divergence is not optional. Taking brainstorming seriously gave us better concepts and better decisions [7]. If I did this project again, I would keep the same process mindset but start earlier and manage time better so that we can allocate adequate time to all of these crucial stages.

CTMFs Used

Freewriting

DIVERGE

Freewriting is my primary diverging tool for generating a large quantity of concepts without filtering too ear...

[View full entry →](#)

NGOs Framework

FRAME

The NGOs Framework (Needs, Goals, Objectives) is a framing tool that translates a broad problem into three lev...

[View full entry →](#)

Proxy Testing

CONVERGE

Proxy testing uses practical substitute measurements when direct real-world testing is not yet feasible. The g...

[View full entry →](#)

DESIGN PROJECT

Motor Fault Detection via Sound Analysis for Professional FPV Drone Operations

SYSTEMS DESIGN

CONTEXT

Worked with D.Vision Aerials to rescope a broad drone diagnostics opportunity into a buildable motor-health solution, then designed and tested a sound-analysis prototype that balanced feasibility, repeatability, and diagnostic value within first-year constraints.

DETAILS

TYPE **Systems Design**
DATE **2026-01 - 2026-04**
CTMFS **3 documented**

TEAM CREDITS

- Collaborative design work completed with my ESC102 Praxis II project team.
- Teammates: Muhammed Ali Bhanji, Michael Lin, and Frank Yin.
- Opportunity framing, NGO development, convergence analysis, and prototype decisions were completed collaboratively.

PROCESS ANNOTATIONS

Summary

In ESC102 Praxis II, our team worked with the professional FPV drone cinematography community in the GTA through interviews and consultation with D.Vision Aerials [10], [14]. The core problem was that pre-flight motor health was being assessed subjectively by ear, even though operations occur in high-risk, high-pressure environments with major liability exposure and tight setup windows [14].

The original framing covered motor, battery, carbon-fiber frame, and radio-frequency diagnostics. After divergence, feasibility checks, and convergence analysis, we narrowed scope to motor fault detection so we could deliver a field-relevant prototype in the available timeline [10], [14]. This let us focus on one high-impact failure pathway while respecting stakeholder constraints around setup speed, repeatability, and non-invasive testing [2]. Early opportunity-space comparisons also included alternate RFP contexts such as ice-removal pathways before final convergence on FPV diagnostics [11], [14].

Outcomes

We designed a pre-flight motor diagnostic prototype that uses sound analysis to support objective go/no-go decisions. The system tests one motor at a time, records audio in a sound-dampening enclosure, computes FFT-based features, and compares them against a healthy baseline profile [10], [12], [15].

The final workflow extracts frequency-domain indicators such as dominant frequency/amplitude, harmonic ratios, significant peak count, and band energy spread. It then classifies each motor into a simplified status output (Ready to Fly, Warning, or Do Not Fly) so results are actionable without requiring raw-signal interpretation on set.

During convergence, we compared this concept against vibration-based analysis (blackbox/IMU) and motor current signature analysis (MCSA). Sound analysis was selected because it had the simplest practical workflow, did not require a test flight, and did not require invasive access to motor terminals [10], [14]. It also provided broader diagnostic coverage in our evaluation (up to 8 fault types, versus 5 for vibration and 1 for current analysis) [15]. Vibration was harder to repeat reliably because each run depended on flight conditions, and current-based analysis was harder to stabilize because battery voltage fluctuations directly shifted the measured current profile.

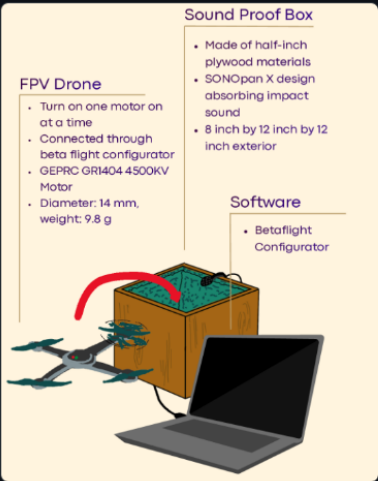


Figure 1: One-pager summary of the opportunity framing, design decisions, and next steps for the motor diagnostic concept [12].

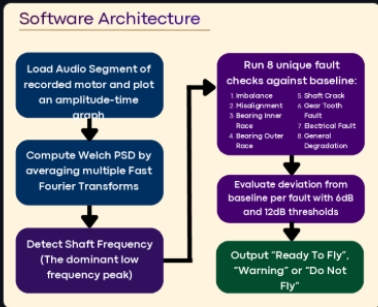


Figure 2: Concept diagram and software workflow used for the sound-analysis motor diagnostic prototype [13].

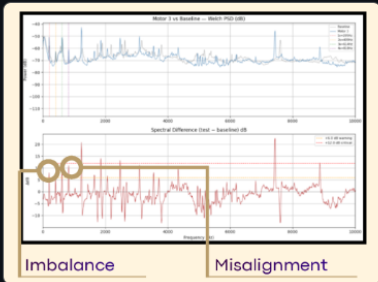


Figure 3: Verification outputs showing spectral indicators used to flag likely imbalance and misalignment behavior [15].

Key outcomes included:

- Rescoped a broad pre-flight diagnostics opportunity into a buildable motor-health problem with explicit needs, goals, and measurable objectives [10], [14]
- Built and demonstrated a working sound-analysis prototype integrating enclosure design, acquisition workflow, FFT feature extraction, and 3-state diagnostic output [10], [12]
- Used comparative convergence tools (measurement matrix + multi-reference Pugh analysis) to evaluate sound, vibration, and MCSA pathways before selecting a final direction [10], [14]
- Verified that the integrated app and diagnostic pipeline could run end-to-end within the project's practical time constraints [10], [15]
- Identified the main remaining gap as fault-verification rigor, with future work focused on larger labeled datasets and stronger repeatability/fault-injection testing [15]

Process Annotations

The biggest turning point for me was when our final app worked end-to-end and integrated all of our ideas into a single prototype. That moment made the project feel real. Up to that point, a lot of our work was still fragmented across separate analyses and partial tests, but once the full pipeline ran cleanly, we could finally evaluate the concept as an actual tool rather than disconnected parts [10], [15].

The hardest part of the process was trying to validate the current-analysis route through simulation. Hardware and software integration was much more difficult than expected, and we eventually found a practical limitation in our Arduino setup, which was not powerful enough for the level of acquisition and processing we needed. That forced a key tradeoff decision: avoid both over-simplifying the problem and over-engineering beyond first-year scope [15]. We had to choose a level of complexity that still produced a usable product while remaining buildable within course constraints [2], [10].

My biggest contribution was building the final prototype itself, especially constructing the sound-dampening box and developing the software workflow that interfaced with the drone and ran diagnostics [10], [12]. The final solution still fell short in one important area: fault-detection confidence. While secondary research supports the diagnostic logic, proving reliability requires much more rigorous fault-injection and repeatability testing than we could complete in this cycle [15].

If I repeated this project, I would put more structure around team communication from the start, with regular progress checkpoints and explicit follow-up when updates stall. The strongest technical ideas only become useful when the team stays synchronized enough to integrate and verify them early.

CTMFs Used

Prototypes

REPRESENT

Prototypes are tangible representations of a concept used to communicate how a design works and to reveal prac...

View full entry →

Requirements List

CONVERGE

A requirements list is a structured document that captures the specific, measurable criteria a design must sat...

View full entry →

Stakeholder Mapping

FRAME

Stakeholder mapping is a systematic process for identifying all parties affected by or involved in a design ch...

View full entry →

DESIGN PROJECT

CIV102 — Structural Bridge Design

STRUCTURAL ENGINEERING

CONTEXT

Designed, analysed, and built a matboard box-girder bridge through multiple quantified design iterations, balancing buckling risk, shear demand, and material constraints before converging on a buildable final geometry.

DETAILS

TYPE **Structural Engineering**DATE **2025-11**CTMFs **3 documented**

TEAM CREDITS

- This bridge was completed as a team project in CIV102.
- Team members: Aditi Datla, Samantha Chang, Perry Xu.
- Analysis, construction, and testing activities were distributed across the team and integrated into a shared final design.

PROCESS ANNOTATIONS

Summary

CIV102 (Structures and Materials) culminates in a bridge design project where teams design and construct a matboard bridge under strict geometric and material constraints [3]. The bridge had to span a 1200 mm support distance (minimum built length 1250 mm), use only the provided matboard and contact cement, and carry a moving train load that increased until failure during testing [8].

The work combined structural analysis and physical fabrication. We used shear-force and bending-moment envelopes, thin-plate buckling checks, glue and matboard shear checks, and repeated cross-section redesign to increase the minimum factor of safety while staying buildable [8], [9].

Outcomes

Our final bridge was a thin-walled box-girder style beam, approximately 1256 mm long, with a double-layer top deck, two vertical walls, an internal top spine through the high-moment region, and 16 diaphragms distributed along the span [8]. Diaphragms were spaced quadratically, denser near supports where shear demand was higher and wider near midspan to reduce unnecessary material.

The CAD model (Figure 1) helped us lock geometry, diaphragm strategy, and splice planning before fabrication, and the constructed bridge (Figure 2) shows how closely the final build followed that design intent [8], [9]. The design process was highly iterative. We moved through major cross-section families (Design 0, modified Design 0, I-section variants, U- and box-like variants, and two Pi-symbol iterations), using calculations and code output each time to identify governing failure modes and decide what to change next [8].

Key design decisions included:

- converging to a box-like/Pi-style cross-section to reduce torsional vulnerability and improve stiffness efficiency relative to I-section concepts
- prioritizing balance between compressive and shear-buckling factors of safety instead of maximizing one metric at the expense of others
- using quadratic diaphragm spacing and planned splice locations to reinforce high-demand regions while preserving material for the full build
- using a double-layer top deck and an internal spine to improve local stability in the most critical bending region



Figure 1: CAD Model of the Final Bridge Design [8]



Figure 2: Final Bridge Built [9]

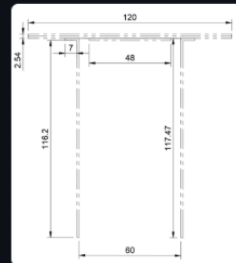


Figure 3: Dimensions of Bridge Cross Section in mm with spine spanning 355mm across center of bridge [8]

Key outcomes included:

- Produced a complete, buildable bridge geometry with matching engineering drawings, cut layout, splice plan, and documented construction process [8], [9]
- Final converged iteration used a 120 mm by 118 mm Pi-style cross-section with 16 diaphragms and balanced critical FOS values (including compressive and local shear-buckling checks) [8]
- Analysis and coding workflow supported evidence-based elimination of weaker iterations and convergence on a design that met project constraints [8]
- Team construction process translated the analytical design into a physical bridge using controlled gluing, staged drying/clamping, and documented quality-control adjustments [9]

Process Annotations

What I am most proud of is our final build quality with limited resources. Our bridge failed earlier than predicted because we underestimated splice weakness and did not reinforce that region enough, but outside of that oversight, the build quality was strong and the overall geometry performed as intended [8], [9].

The biggest turning point was fixing our matboard cut-layout problem. A measurement error made it look like the bridge would not fit without major compromises, which caused a lot of panic. Once Perry found workable tradeoffs using Aditi's optimization algorithm, we regained confidence and moved forward with a design that was both buildable and promising [8], [9].

Early on, we eliminated Design 0-adjacent concepts, especially base-heavy options, because modelling showed they used material inefficiently: tensile capacity became too strong while compression still governed [8]. The key tradeoff was compression versus shear, so we tuned height-to-thickness ratios to balance both in code. The hardest part was then executing the build, which took about 20 hours of sustained precision [9]. That made one lesson clear for me: a theoretically stronger bridge is not better if it is too difficult to build cleanly. If I repeated this project, I would start cross-section optimization earlier and enforce stricter team cross-checking, because even strong contributors are human and early verification prevents costly mistakes.

CTMFs Used

Iterative Design

DIVERGE

Iterative design is a cycle of proposing, evaluating, and refining design options over multiple passes. Instea...

[View full entry →](#)

Mathematical Modelling

CONVERGE

Mathematical modelling applies equations and engineering assumptions to predict behavior before physical testi...

[View full entry →](#)

Technical Drawing

REPRESENT

Technical drawing is the practice of creating precise, standardised visual representations of a design — inclu...

[View full entry →](#)

DESIGN VOCABULARY

Concepts, Tools, Models & Frameworks

The CTMFs I've applied across my design projects, organised by strand:
Frame, Diverge, Converge, and Represent. Filter by strand or project.

STRAND

All

Frame

Diverge

Converge

Represent

PROJECT

All

CIV102

Roller Support Design to Prevent Smudging for Left-Handed Writing in the EngSci Common Room

Motor Fault Detection via Sound Analysis for Professional FPV Drone Operations

Showing 9 of 9 CTMFs

Freewriting

DIVERGE

Freewriting is my primary diverging tool for generating a large quantity of concepts without filtering too early. I use it to forc...

PRAXIS-I

Iterative Design

DIVERGE

Iterative design is a cycle of proposing, evaluating, and refining design options over multiple passes. Instead of committing to o...

CIV102-BRIDGE

Mathematical Modelling

CONVERGE

Mathematical modelling applies equations and engineering assumptions to predict behavior before physical testing. In design conver...

CIV102-BRIDGE

NGOs Framework

FRAME

The NGOs Framework (Needs, Goals, Objectives) is a framing tool that translates a broad problem into three levels: what users actu...

PRAXIS-I

Prototypes

REPRESENT

Prototypes are tangible representations of a concept used to communicate how a design works and to reveal practical issues before ...

PRAXIS-II

Proxy Testing

CONVERGE

Proxy testing uses practical substitute measurements when direct real-world testing is not yet feasible. The goal is to answer spe...

PRAXIS-I

Requirements List

CONVERGE

A requirements list is a structured document that captures the specific, measurable criteria a design must satisfy. Requirements a...

PRAXIS-II

Stakeholder Mapping

FRAME

Stakeholder mapping is a systematic process for identifying all parties affected by or involved in a design challenge, then visual...

PRAXIS-II

Technical Drawing

REPRESENT

Technical drawing is the practice of creating precise, standardised visual representations of a design — including dimensioned ort...

CIV102-BRIDGE

CTMF ENTRY

Freewriting

DIVERGE

Used in: [praxis-1](#)

01 Explanation

Freewriting is my primary diverging tool for generating a large quantity of concepts without filtering too early. I use it to force ideation without the limitation of implementation, make hidden assumptions obvious, and capture partial thoughts that can later be refined into testable concepts [1], [7].

02 Evidence of Use

In Praxis I, our freewriting rounds produced many concepts before convergence, including quick-drying ink, a dummy-marker, air-cushion or fan-assisted support, marker-mounted platforms, and rolling support mechanisms. As shown in Figure 6, the freewriting output contains a mix of feasible and exploratory ideas, which is what made it valuable for early divergence. Across our diverging methods, freewriting consistently generated ideas that were most reusable in later concept development [6], [7].

03 Assessment of Utility

Freewriting was most useful for breaking the initial cognitive anchor and giving our team enough concept diversity to compare meaningfully during convergence. It also exposed early technical questions and pointed us toward reference designs faster than discussion-only ideation [6], [7]. Its main limitation is that raw output quality varies, so it must be followed by a round of filtering through discussion and secondary research rather than treated as decision evidence on its own.

04 Fit with My Practice

This fits my position of widening the space first, then letting evidence drive the next decision. Strong concepts emerge when time and effort is given to diverging, but freewriting still needs to be filtered with convergence tools so quantity is not prioritized above quality [6], [7].

Kind of plausible ideas (Inspired by Wishing Diverging Method and free writing)

1. Improved material (stain + sweat resistance) for the glove so that there is no smear at all (whatever the criteria is for an improvement on the smearing)
2. Quick drying markers that requires special eraser to be erased so hands can't erase the marker -> change the chemical composition of the ink
 - a. Reference design: Eraseable pens
3. Repelling magnet -> magnetic levitation
 - a. Could also use electromagnets
4. Air cushion attachment
 - a. Get exact values later from reputable sources but roughly the following:
 - i. The average person exert 1.5N of force on the table while writing at a normal speed
 - ii. A rough estimate of the area of contact is 30cm^2
 - iii. A rough pressure exerted by the hand is 50 Pa
 - iv. Figure out how much air flow rate is needed to exert an equivalent opposite pressure of 50 Pa
 - b. The fan cannot be too loud nor can the attachment be too heavy
 - c. Porous surface at the bottom for which the air can go through
5. Plastic attachment to hand to contact board
 - a. Rolling ball?
6. Digital board (can just busy)
7. Make a supporting section/anchor/platform on the marker that the palm rests on instead of the board.
8. Bent marker that writes to the left of it instead of where it's pointed?
 - a. Can be held just like a normal marker
 - b. Solution one: over the hand attachment
 - c. Solution two: around the hand attachment
9. Dummy marker (basically have a second marker to the left of the real one that is the one that actually draws)

Figure 6: Freewriting excerpt from Praxis I Alpha Release brainstorming notes [7].

CTMF ENTRY

Iterative Design

DIVERGE

Used in: `civ102-bridge`

01 Explanation

Iterative design is a cycle of proposing, evaluating, and refining design options over multiple passes. Instead of committing to one concept immediately, the team improves the design through repeated adjustments informed by feedback and analysis [3], [8].

02 Evidence of Use

In CIV102, our bridge design evolved through repeated revisions to geometry and member layout as we compared alternatives and checked expected performance. We used each iteration to identify governing failure modes and decide what to change next, instead of treating the first concept as fixed [8], [9].

03 Assessment of Utility

Its main utility was reducing early design attachment. Iteration allowed us to improve weaker parts of the bridge concept before finalizing construction details [8], [9]. It allowed us to identify what aspects of a current iteration are strongest while being able to differentiate the weaknesses, thus giving a direction to move forward with.

04 Fit with My Practice

This fits my position because I work by breaking complex systems into manageable parts, testing

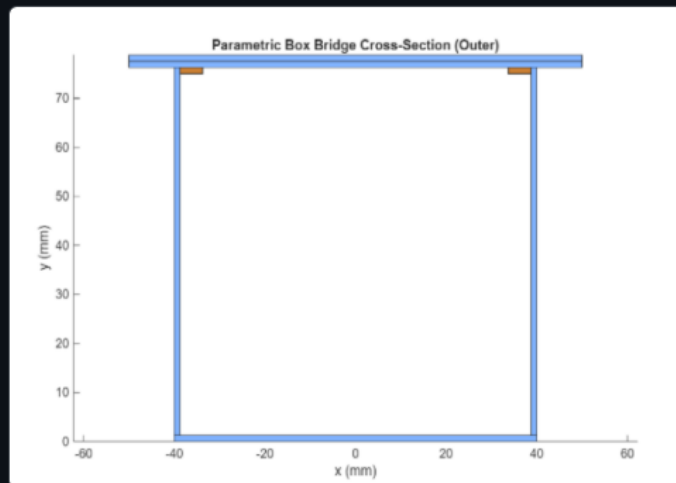


Figure 1: Cross-section of the second iteration, modified to maximize crucial properties, such as the value of the second moment of area and the neutral axis. Design 0 Modified solidified the elimination of a design akin to Design 0 due to its vulnerability to compressive forces [8].

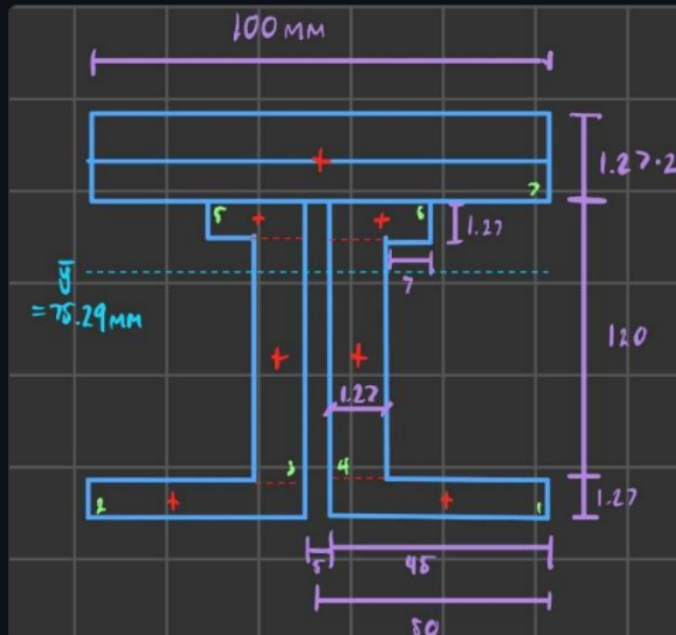


Figure 2: Takes inspiration from a traditional I-Beam but splits it at the leg, however resulted in too much wasted material for not enough strength, thus discarding this cross section [8].

84 Fit with My Practice

This fits my position because I work by breaking complex systems into manageable parts, testing behavior, and then adjusting based on what the evidence shows. Iterative design mirrors that exact approach: each bridge revision became a focused test of one assumption, then informed the next change. It also keeps me from becoming attached to an idea when there are many other things to be verified, such as constructability [8], [9] [4].

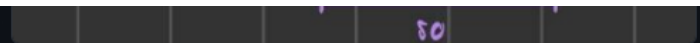


Figure 2: Takes inspiration from a traditional I-Beam but splits it at the leg, however resulted in too much wasted material for not enough strength, thus discarding this cross section [8].

CTMF ENTRY

Mathematical Modelling

CONVERGE

Used in: [civ102-bridge](#)

01 Explanation

Mathematical modelling applies equations and engineering assumptions to predict behavior before physical testing. In design convergence, it helps compare options quantitatively and justify decisions [3], [8].

02 Evidence of Use

In CIV102, we used structural calculations based on loading and section behavior to estimate critical regions and expected failure tendencies before real life testing. These model-based predictions informed our design choices and where to reinforce the bridge [8]. Figure 1 shows the modelling output we used to compare load effects and decide where reinforcement was necessary.

03 Assessment of Utility

Mathematical modelling improved confidence in our design decisions by providing quantitative guidance before construction. It narrowed the solution space and reduced purely trial-and-error adjustments [8], [9].

However, it is not perfect and can sometimes fail to account for a failure mode. For example, our bridge failed due to the splice being weaker than expected and the mathematical model we created did not account for it.

04 Fit with My Practice

This fits my position because I work by breaking complex systems into manageable parts, testing assumptions, and adjusting from evidence. Mathematical modelling supports that directly: it lets me isolate one variable at a time, check expected behavior before building, and only commit once the numbers and construction implications are both credible [8], [9].

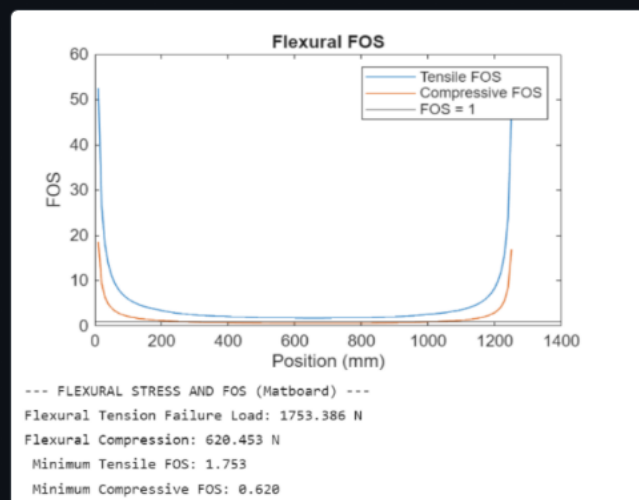


Figure 1: Mathematical modelling output used to compare load effects and support reinforcement decisions before fabrication [8].

CTMF ENTRY

NGOs Framework

FRAME

Used in: `praxis-1`

01 Explanation

The NGOs Framework (Needs, Goals, Objectives) is a framing tool that translates a broad problem into three levels: what users actually need, what the design should achieve, and how success will be measured. It helps prevent jumping to solutions by keeping design decisions tied to stakeholder outcomes [1], [5].

02 Evidence of Use

In Praxis I, we used NGOs while rescoping the adopted whiteboard brief. We reframed the core need as preventing smudging and ink transfer for left-handed writers, set goals around preserving natural writing and comfort, and defined measurable objectives for writing-speed retention, low visible smudging, ergonomic wrist posture, and acceptable mass [5], [6].

03 Assessment of Utility

NGOs made convergence more rigorous because every concept could be tested against the same objective criteria instead of preference. It also helped eliminate concepts that performed well in one area but failed the broader design intent [6].

04 Fit with My Practice

This fits how I approach engineering design because it lets me break a problem into core functionalities and tackle each one individually. Once each part is defined and working on its own, I can focus on integration and isolate issues more clearly when they appear [5], [6].

CTMF ENTRY

Prototypes

REPRESENT

Used in: `praxis-ii`

01 Explanation

Prototypes are tangible representations of a concept used to communicate how a design works and to reveal practical issues before final implementation. They can be low- or high-fidelity depending on the question being answered [2], [10].

02 Evidence of Use

In Praxis II, our team built multiple prototype iterations to represent the converged concept and evaluate how design changes affected usability and performance. Figure 1 shows the one-pager visual summary used to communicate the concept, prototype direction, and iteration outcomes during team reviews and stakeholder-facing discussions [10], [12], [13].

03 Assessment of Utility

Prototyping improved design quality by exposing issues that were not obvious in discussion alone. It also made team decisions faster because alternatives could be compared based on observed behavior rather than assumptions [10], [15].

04 Fit with My Practice

This fits my position because I understand systems best by testing behavior rather than trying to design everything perfectly up front. Prototyping lets me verify core functions first, adjust quickly, and then integrate with more confidence once each part has been tested under realistic conditions [10], [15].

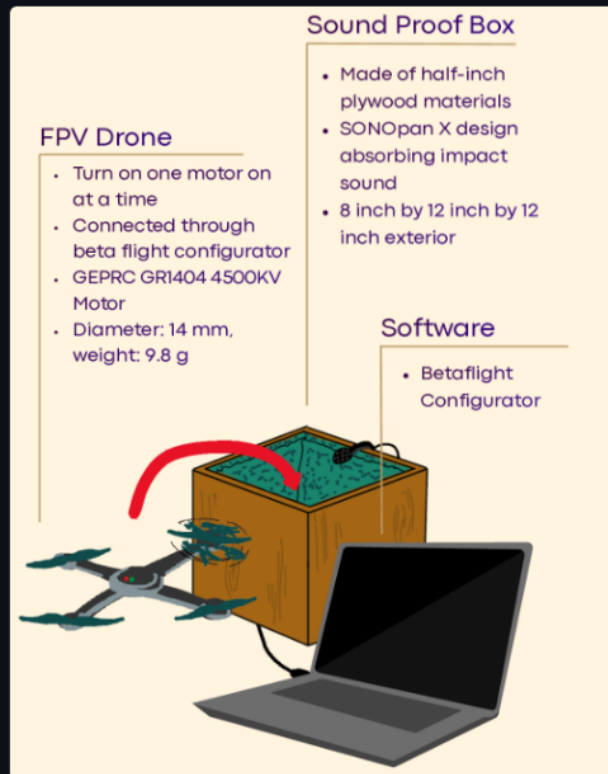


Figure 1: Praxis II one-pager used to present prototype direction, key design choices, and iteration outcomes [12].

CTMF ENTRY

Proxy Testing

CONVERGE

Used in: [praxis-i](#)

01 Explanation

Proxy testing uses practical substitute measurements when direct real-world testing is not yet feasible. The goal is to answer specific performance questions with the simplest credible method under project constraints [1], [6].

02 Evidence of Use

In Praxis I, the team used proxy tests to compare concept performance under first-year constraints: writing speed versus a control condition, smudging through color-difference comparison, wrist-angle ergonomics using a modified RULA-style check, and concept mass checks. As shown in Figure 7, these proxy methods converted broad concept claims into directly comparable evidence across roller support, dummy marker, platform, and air-cushion concepts [6], [7].

Proxy Test D1: Impact of Prototypes on Writing Speed

This proxy verifies requirement 1 (design cannot inhibit user's baseline DASH-2 score) by testing writing speed with the 3 concepts, the dummy marker, the platform and the roller support. The roller support is tested by substituting for 0.4kg wrist weights to simulate the extra weight of the roller support. This weight is an overestimate for the roller support but if it is able to pass this max threshold as defined by requirement 1, then a lighter design passes too. This substitution also works because the roller support is attached to the wrist and not to the marker, allowing the user to write in their natural position.

1. The control test defines a mean baseline writing speed for a left-handed writer writing on a whiteboard with no additional weight or using a normal marker
2. The sentence given is "The quick brown fox jumps over the lazy dog" because it contains every letter of the alphabet (pangram) and is short enough to avoid long writing session fatigue
3. Each prototype is tested, collecting a mean of 5 times to reduce error, with the 0.4 kg wrist weights used for the roller support
 - a. Writing speed is considered hindered if the mean time with added weight has a z-score greater than 2 (95% confidence interval) relative to the control mean. Therefore we can establish the effect of the given weight in hindering writing speed with a significance value of $\alpha = 0.05$

Figure 7: Example proxy test from Praxis I used to compare concept performance under constrained testing conditions [7].

03 Assessment of Utility

Proxy testing made convergence evidence-based instead of preference-based. It exposed tradeoffs clearly and helped justify why the roller concept was the strongest overall option rather than just the most intuitive one [6].

04 Fit with My Practice

This fits my position of testing assumptions early and making evidence-based decisions under real constraints. Convergence became stronger when tradeoffs were tested explicitly, especially when speed results showed why roller support was preferable to dummy marker [6].

CTMF ENTRY

Requirements List

CONVERGE

Used in: [praxis-ii](#)

01 Explanation

A requirements list is a structured document that captures the specific, measurable criteria a design must satisfy. Requirements are typically categorised as functional (what the design must do), non-functional (qualities it must have), and constraints (boundaries it must respect). Well-written requirements are testable, traceable to stakeholder needs, and prioritised [2], [14].

02 Evidence of Use

In Praxis II, our team used a weighted requirements list linked to stakeholder needs and tradeoffs. We used it to compare concept directions, justify convergence decisions, and evaluate prototype iterations against consistent criteria rather than ad-hoc preferences [10], [14].

03 Assessment of Utility

This was one of the most useful convergence tools because it connected stakeholder framing to measurable decision-making. Once criteria and weightings were explicit, team discussions became faster and more defensible [10], [15].

04 Fit with My Practice

This fits my position because it turns broad concerns into testable criteria and keeps scope credible. Explicit requirements helped avoid over-engineering by focusing only on what could be verified and delivered within the project's constraints [2], [10], [15].

CTMF ENTRY

Stakeholder Mapping

FRAME

Used in: `praxis-ii`

01 Explanation

Stakeholder mapping is a systematic process for identifying all parties affected by or involved in a design challenge, then visualising their relationships, interests, influence, and potential conflicts. Common representations include influence-interest matrices and network maps [1], [2].

02 Evidence of Use

In Praxis I, stakeholder mapping helped us separate primary users (left-handed board writers) from secondary and tertiary stakeholders such as audience readers and facilities staff, which clarified what counted as success beyond just user comfort [6]. In Praxis II, we expanded this process for the FPV context by mapping pilots, engineers, operations/compliance, production clients, bystanders, regulators, and insurers, then explicitly identifying conflicts between speed, safety, and accountability before converging on a scoped solution [10], [14].

03 Assessment of Utility

Highly useful as a starting point for any design project. It forces you to look beyond the obvious 'user' and consider the full ecosystem a design operates within. The structured visualisation makes implicit assumptions explicit, which is especially valuable in team settings where different members may have different mental models of who matters [10], [14].

04 Fit with My Practice

This fits how I work because I usually break complex problems down before trying to integrate everything. Stakeholder mapping helps me identify which users and constraints need to be handled first, so I can prioritize decisions properly and isolate issues instead of treating everything as equally urgent [10], [14].

CTMF ENTRY

Technical Drawing

REPRESENT

Used in: `civ102-bridge`

01 Explanation

Technical drawing is the practice of creating precise, standardised visual representations of a design — including dimensioned orthographic views, cross-sections, and detail drawings. These drawings communicate exact geometry, materials, tolerances, and assembly information to anyone who needs to build, evaluate, or modify the design [3], [9].

02 Evidence of Use

For the CIV102 bridge project, I created detailed technical drawings including plan, elevation, and cross-sectional views with dimensions and assembly details. These drawings were used to communicate geometry clearly before construction and to keep team fabrication aligned with the analysed design [8], [9]. Figure 1 is the final drawing package view that the team used as the shared build reference.

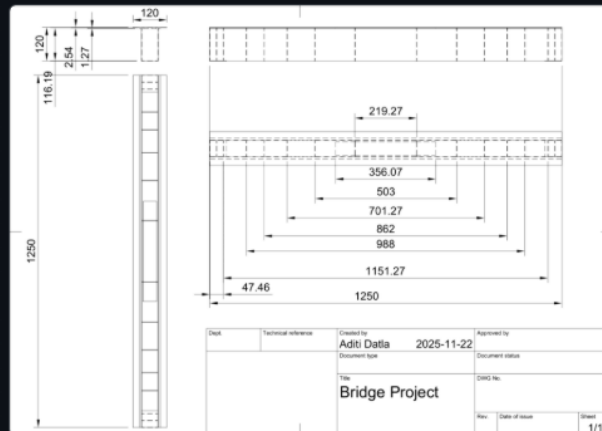


Figure 1: Final Bridge Technical Drawing [8]

03 Assessment of Utility

Essential for any design that needs to be physically constructed or manufactured. Technical drawings impose a discipline of precision that catches errors — you can't dimension something without deciding exactly how big it is, which often reveals decisions you haven't actually made yet [8], [9].

04 Fit with My Practice

This fits my practice because technical drawings force decisions to be explicit before fabrication, which supports my evidence-first approach. It also connects to my CIV102 reflection on build quality and team cross-checking: clearer drawings reduce ambiguity and prevent avoidable construction errors [8], [9].

ABOUT

About Me

I am a first year Engineering Science student at the University of Toronto, with a passion for engineering design and curiosity across several disciplines. This portfolio is a way for me to document my engineering journey, share the projects I have worked on, reflect on the design process, and demonstrate how I think about engineering challenges.

What I'm Working On

My current focus is to build strong fundamentals in core engineering concepts while attempting to explore my interests, in hopes that I can find my passion. Despite being pulled towards Machine Intelligence and Robotics, I want to give myself the opportunity to explore other disciplines.

Outside the classroom, I'm interested in projects that integrate hardware and software, such as my participation with U of T Formula Racing (UTFR) within the driverless division, my Praxis II project which involved drone diagnostics and more. I am greatly invested in these types of projects because I believe some of the most interesting engineering challenges emerge when you have to make physical and digital systems work together.

Beyond Engineering

When I'm not studying or working on projects, you can find me exploring new technologies, diving into whatever rabbit hole has captured my attention that week, or even just enjoying some games to take my mind off work. I believe that the best engineers are broadly curious people, letting that curiosity to guide me

QUICK INFO

PROGRAM	Engineering Science
YEAR	First Year (EngSci)
UNIVERSITY	University of Toronto

INTERESTS

- Machine Learning/AI
- Robotics
- Embedded Systems
- Hardware and Software Integration
- Human-Centred Design

TECHNICAL SKILLS

Technologies and tools I work with

Languages

- Python
- C++
- C
- Java
- C#
- MATLAB

Tools & Platforms

- Git
- Docker
- Google Cloud
- Jira
- Slack
- Raspberry Pi
- RDS
- Fusion 360
- Microsoft Office

Libraries & Frameworks

- FastAPI
- NumPy
- OpenCV
- TensorRT
- NVIDIA DeepStream
- MediaPipe
- React

BIBLIOGRAPHY

References

References cited throughout this portfolio in IEEE format, using the Praxis embedded extract convention — each entry includes the citation followed by the context in which the source was used.

-
- 01 [1] University of Toronto, "ESC101 20259 Praxis I assignment documentation set (Process Analysis, Design Brief, Alpha Release, and Design Report documents with associated IATs)," course handouts and assessment documents, 2025.

EMBEDDED CONTEXT: Primary assignment-level reference set used across Praxis I project summaries and CTMF evaluations.

- 02 [2] University of Toronto, "ESC102 20261 Praxis II assignment documentation set (Student Engineer Design Portfolio brief and IAT)," course handouts and assessment documents, 2026.

EMBEDDED CONTEXT: Primary assignment-level reference set used for Praxis II portfolio framing, CTMF structure, and evaluation criteria.

- 03 [3] University of Toronto, "CIV102 bridge assignment documentation set (bridge project handout and team deliverables)," course assignment materials, 2025.

EMBEDDED CONTEXT: Assignment-level source set used throughout the CIV102 bridge project and associated CTMF entries.

- 04 [4] M. H. Heryani, "Position Statement Video," YouTube video, 2026.

EMBEDDED CONTEXT: Video source referenced in the Position Statement page as primary evidence of design-position articulation.

- 05 [5] Praxis I Team, "Rescoped Design Brief," ESC101 Praxis I, 2025.

EMBEDDED CONTEXT: Primary Praxis I framing source used for opportunity interpretation, scope refinement, and requirement direction.

- 06 [6] Praxis I Team, "Design Report (Roller Support to Prevent Left-Handed Whiteboard Smudging)," ESC101 Praxis I, 2025.

EMBEDDED CONTEXT: Primary Praxis I evidence source for convergence logic, requirement outcomes, testing summaries, and final recommendation.

- 07 [7] Praxis I Team, "Alpha Release Brainstorming," ESC101 Praxis I, 2025.

EMBEDDED CONTEXT: Primary evidence source for Praxis I diverging methods, concept generation, and pre-convergence records.

08

[8] Team 509, "CIV102 Project Team 509 Design Report," course project report, 2025.

EMBEDDED CONTEXT: Primary technical source for CIV102 bridge analysis choices, iteration history, and final design rationale.

09

[9] Team 509, "CIV102 Project Team 509 Engineering Assembly," course project assembly documentation, 2025.

EMBEDDED CONTEXT: Primary construction and assembly source for fabrication sequence, quality control, and final build evidence.

10

[10] Praxis II Team, "Beta release document," ESC102 Praxis II, 2026.

EMBEDDED CONTEXT: Source for Praxis II beta-stage prototype integration, decision records, and readiness evidence.

11

[11] Praxis II Team, "RFP - Ice Removal" (project context document), ESC102 Praxis II, 2026.

EMBEDDED CONTEXT: Used as contextual evidence for early opportunity-space comparison in Praxis II before final scope convergence.

12

[12] Praxis II Team, "One Pager (Brochure)," ESC102 Praxis II, 2026.

EMBEDDED CONTEXT: Primary communication source for final concept summary and stakeholder-facing project representation.

13

[13] Praxis II Team, "Poster," ESC102 Praxis II, 2026.

EMBEDDED CONTEXT: Primary visual communication source for condensed presentation of project evidence and conclusions.

14

[14] Praxis II Team, "RFP I - FPV Drone Diagnostics," ESC102 Praxis II, 2026.

EMBEDDED CONTEXT: Primary requirements and context source for FPV diagnostics framing and stakeholder-aligned scoping.

15

[15] Praxis II Team, "Verification Processes," ESC102 Praxis II, 2026.

EMBEDDED CONTEXT: Primary verification-method source for diagnostics testing logic and evidence of prototype assessment.

16

[16] Praxis I Team, "Opportunity Scoping," ESC101 Praxis I, 2025.

EMBEDDED CONTEXT: Supporting source for early opportunity selection and narrowing decisions in Praxis I.

17

[17] Praxis I Team, "Pairwise Comparison Matrix Arguments," ESC101 Praxis I, 2025.

EMBEDDED CONTEXT: Supporting convergence source used to justify tradeoffs between competing concept directions in Praxis I.

[5] Praxis I Team — Rescoped Design Brief

Extract:

mean.

VI. Needs, Goals and Objectives

From the stakeholder expectations and reference design shortcomings, we can derive needs, goals and objectives necessary to deem a solution to this *splartz* “sufficient.”

Need: Left-handed students need ~~are able~~ to write on whiteboards in the EngSci common room without smudging their hands or hindering their normal writing ability. ~~issue~~.

Table 1: Goal 1 meets primary stakeholders’ needs by facilitating their use of whiteboards without harm.

	Description	Metric	Justification
Goal 1	Design must not affect the user’s baseline writing ability		
Requirement 1.1	Cannot inhibit the user’s baseline DASH-2 score by more than 15 points [16].	Score (points) on the DASH-2 (Detailed Assessment of Speed of Handwriting).	A 15-point decrease on the DASH-2 is one standard deviation, signifying a significant drop in handwriting speed [16].
Evaluation Criteria 1.1	The fewer points inhibited by the design on the user’s baseline DASH-2 score the better.		

Commented [SZ1]: Rewrote to be more specific about left-handed EngSci students’ exact needs.

Requirement 1.2	Ensure there is no risk of strain injury caused by wrist extended by $>90^\circ$, wrist radial abduction, $>20^\circ$ and wrist ulnar abduction, $>30^\circ$, outlined in ISO 11226-2000 [Appendix A] [13].	Angle of extreme wrist extension, wrist radial abduction and wrist ulnar abduction in degrees.	Reducing strain injuries when adopting a standing stance re-ensures the user’s original writing ability.
Evaluation Criteria 1.2	The greater range of angles less than 90° , 20° , and 30° for wrist extension, wrist radial abduction, and wrist ulnar abduction respectively that the design accommodates for the better.		Greater ranges of angles accommodate for more people’s unique grip styles
Requirement 1.3	Accommodate for at least 86% of the population by allowing various grip styles on the pen, namely lateral tripod, dynamic tripod, lateral quadropod, and dynamic quadropod. Accommodate for various grip styles on the pen, namely tripod and quadropod grip styles.	Grip styles: Lateral tripod, dynamic tripod, lateral quadropod, dynamic quadropod.	Ensures the design is usable by any users with differing grip styles [14]. D/Ergonomics [27] emphasises to “anticipate actions.”
Evaluation Criteria 1.3	The more grip styles the pen can accommodate the better.		

Commented [MZZ]: Accommodating for the 4 mentioned grip styles ensures at least 86% of users 11 years and older will be able to hold the product the same way they normally write.
https://www.researchgate.net/publication/311881226_Pen_Grip_Patterns_of_Students_and_Skilled_Workers

Commented [MZZ]: A small number of users may exhibit an unusual grip style that differs significantly from the common grip styles.

Table 2: Ergonomics-focused goal that meets the needs of all primary stakeholders through ensuring the device is convenient to use and be around.

	Description	Metric	Justification
Goal 2	Design must be comfortable to use and have around the commons.		
Requirement 2.1	Design width of at most 4.9cm—Design can rest on a 4.9cm and 5.5cm marker tray with a velocity of 0 m/s in any direction.	Width (cm) of design; Velocity (m/s) of design when it rests on a marker tray.	Design must fit onto the marker trays (4.9cm and 5.5cm) so users can have quick access.

Supports: This extract supports the Praxis I framing claim by showing how the problem was defined around left-handed smudging and translated into measurable design direction.

Extract:

6 Recommended Roller Support Design

This section will describe the recommended design, justify key design decisions about it, and provide an overview of why it is the recommended design.

6.1 Roller Support Design Description

The design that we converged onto was a ball support on a wrist cuff, as shown in Figure 3. Users hold a marker normally in their hand and their wrist/arm is supported by the platform attached to the ball contacting the whiteboard. The ball does not smudge noticeable amounts of ink on the whiteboard because if it doesn't slip, it only pushes ink into the whiteboard. [\[Video\]](#).



Figure 4: A ball transfer unit has many low-friction small balls surrounding the main ball [14]. This lets the main ball spin freely while attached to the arm support platform.

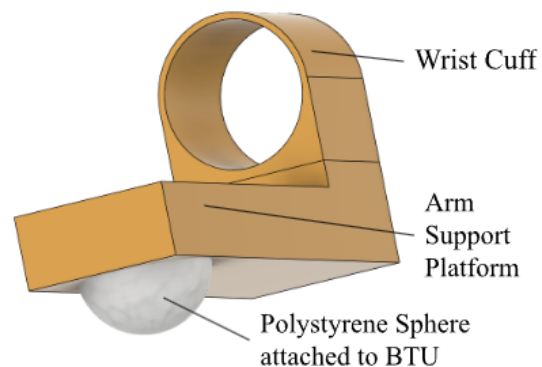


Figure 3: Roller support design. The arm rests on the flat platform and the wrist goes through the wrist cuff. The sphere is attached to the ball transfer unit (BTU) as shown in Figure 4.

Supports: This extract supports convergence and outcome claims by showing the final concept selection from the final submitted design report

7 Praxis I Team — Alpha Release Brainstorming

Extract:

Kind of plausible ideas (Inspired by Wishing Diverging Method and free writing)

1. Improved material (stain + sweat resistance) for the glove so that there is no smear at all (whatever the criteria is for an improvement on the smearing)
2. Quick drying markers that requires special eraser to be erased so hands can't erase the marker -> change the chemical composition of the ink
 - a. Reference design: Eraseable pens
3. Repelling magnet → magnetic levitation
 - a. Could also use electromagnets
4. Air cushion attachment
 - a. Get exact values later from reputable sources but roughly the following:
 - i. The average person exert 1.5N of force on the table while writing at a normal speed
 - ii. A rough estimate of the area of contact is 30cm^2
 - iii. A rough pressure exerted by the hand is 50 Pa
 - iv. Figure out how much air flow rate is needed to exert an equivalent opposite pressure of 50 Pa
 - b. The fan cannot be too loud nor can the attachment be too heavy
 - c. Porous surface at the bottom for which the air can go through
5. Plastic attachment to hand to contact board
 - a. Rolling ball?
6. Digital board (can just busy)
7. Make a supporting section/anchor/platform on the marker that the palm rests on instead of the board.
8. Bent marker that writes to the left of it instead of where it's pointed?
 - a. Can be held just like a normal marker
 - b. Solution one: over the hand attachment
 - c. Solution two: around the hand attachment
9. Dummy marker (basically have a second marker to the left of the real one that is the one that actually draws)

Absurd ideas (Wishing Diverging Method):

1. I wish people would just write with the right hand
2. I wish I could change english to be written from right to left and force all righties to write with left hand
3. I wish I could invert the physical properties of the universe so that left now becomes right for specific people
4. I wish I could write a sentence from the end to the start
5. Reverse time locally on the board/delay time on the whiteboard
6. Biomedical engineering so that all babies are born right handed/ambidextrous
7. Marker that communicates with a second robot marker. The real marker won't draw anything, but it'll tell the robot marker to draw that thing a few seconds after the real marker draws it -> maybe a dummy marker physically attached instead of wirelessly connected?
8. Hire another person to draw for you
9. Make schools to force left handed people to write right handed
10. Transfer what wants to be drawn in brainwaves to the whiteboard

Supports: This extract supports divergence claims by showing early freewriting/brainstorming outputs with multiple distinct concept paths before convergence.

8 Team 509 — CIV102 Design Report

Extract: Figures found on following pages: Technical Drawing, Mathematical Modelling, Iterative Design, CIV102 – Bridge Project Page

Supports: This extract supports analysis and iteration claims by showing structural reasoning used to compare cross-sections and justify the final design direction.

9 Team 509 — CIV102 Engineering Assembly

Extract: Figures found on following pages: Technical Drawing, Mathematical Modelling, Iterative Design, CIV102 – Bridge Project Page

Supports: This extract supports constructability claims by showing fabrication/assembly decisions and how the analytical design was translated into a physical build.

10 Praxis II Team — Beta Release Document

Extract:

Measurement Matrix — NEW VERSION

Status key: Confirmed = meets requirement by inspection or test. Partial = partially meets requirement. Needs testing/research = not yet verified.

Requirement	Sound Analysis	Vibration Analysis	Current Analysis (MCSA)
1.1 # measurable parameters indicative of motor health	4 feature categories: fundamental freq & amplitude; harmonic amplitude ratios; significant peak count; band energy (low/mid/high)	1 parameter: GyroADC angular velocity → FFT amplitude spectrum.	1 parameter: Phase current → FFT fault-frequency components (bearing, winding, demagnetisation sidebands).
1.2 Sensor consistency across ≥5 runs under controlled conditions	Confirmed	Lack of consistency	Confirmed
1.3 Data acquisition TACOM score ≤ 3.6	2.0	2.9	2.5
1.4 Zero irreversible physical operations	Confirmed by inspection	Confirmed Micro USB access only — no hardware modification.	Confirmed reversible operations XT30 pigtailis plug/unplug — no soldering. Invasive location
1.5 Zero external network calls during diagnostic cycle	Confirmed all operations run locally	Confirmed Blackbox export and FFT run locally via USB.	Confirmed ESP32 acquisition and FFT onboard or local serial only.
1.6 ≥1 detectable motor fault	8 fault types detectable	5 fault types detectable	1 fault types
2.1 Per-motor healthy baseline from ≥5 calibration runs	Confirmed Brand new motor used as reference, ≥5	Theoretically 4 brand new motors used as reference, ≥5.	Brand new motor used as reference, ≥5.
2.2 Classification output within 15 s of last data sample	3-4 seconds taken to output results	7.613 s	Nearly instantaneous
2.3 Fault flagged at kσ deviation from baseline	Confirmed the fault is detected at 2 standard deviations away	The fault is detected at 2 standard deviations away	The fault is detected at 3 standard deviations away

2.4 Three-state output: Pass / Warning / Fail	Confirmed fault detection logic implemented as such	Confirmed	Confirmed
3.1 Result communicated without raw data interpretation	The confirmed result is shown as either "Ready To Fly", "Warning" or "Do Not Fly".	Confirmed	Confirmed
5.1 Operates in -10°C to 40°C	Confirmed Lavalier mic typically operates from -30 °C to +85 °C.	Confirmed ICM-42688-P rated -40°C to +85°C. Exceeds requirement.	Confirmed ACS712: -40°C to +150°C. ESP32: -40°C to +85°C. Both exceed requirements.
5.2 Fits within 30 000 cm³	27 000 cm³	Micro usb cable plus existing laptop only. Additional volume is	17.2 cm³

Supports: This extract supports beta-stage implementation and convergence claims by showing the measurement matrix used to evaluate Sound, Vibration, and MCSA concepts against project requirements and confirm readiness decisions

12 Praxis II Team — One Pager (Brochure)

Extract:

MOTOR FAULT DETECTION

via sound analysis

Key Design Decisions

Decision 1 — Brand new motor as healthy baseline

- Generated a healthy FFT baseline to compare against other drone motors

Decision 2 — Enclosure material selection

- Outer shell — half-inch plywood to block noise
- Inner lining — SONOPanX 7/16" fibreboard to absorb sound

The Opportunity

- Professional FPV Drone cinematography drones' motor health is currently checked in a very subjective way
- You hold the drone to your ear, if it sounds smooth, it flies.

Next Steps

- Use thicker enclosure materials
- Create airtight seal on the box
- Replace the single-motor healthy baseline with a large dataset of healthy motor signatures
- Train a machine learning model

UNIVERSITY OF TORONTO D.VISION AERIALS

UNIVERSITY OF TORONTO D.VISION AERIALS

Page 2 - INSIDE

The Concept

FPV Drone

- Turn on one motor on at a time
- Connected through beta flight configurator
- GEPRC GR404 4500KV Motor
- Diameter: 14 mm, weight: 9.8 g

Sound Proof Box

- Made of half-inch plywood materials
- SONOPan X design absorbing impact sound
- 8 inch by 12 inch by 12 inch exterior

Software

- Betaflight Configurator

Software Architecture

```
graph TD
    A[Load Audio Segment of recorded motor and plot on amplitude-time graph] --> B[Compute Welch PSD by averaging multiple Fast Fourier Transforms]
    B --> C[Detect Shift Frequency (The dominant low frequency peak)]
    C --> D[Run 8 unique fault checks against baseline:  
1. Loose Motor  
2. Motor Fault  
3. Motor Fault  
4. Motor Fault  
5. Motor Fault  
6. Motor Fault  
7. Motor Fault  
8. Motor Fault]
    D --> E[Exclude deviation from baseline per fault with fail and pass thresholds]
    E --> F[Output "Ready To Fly", "Warning" or "Do Not Fly"]
```

Why Sound Analysis? Eliminated Designs:

Vibration Analysis (FC Blackbox)

- Requires opening Betaflight and doing a full test flight — too slow for quick pre-flight checks.
- Gyro readings also change a lot between motors, so there's no reliable baseline.

Current Signature Analysis (MCSA)

- Checking phase current requires soldering to the controller board
- Risks damage, takes considerable time

Research-Backed Approach

"Vibration-based Condition Monitoring: Industrial, Aerospace and Automotive Applications by Robert Bond Randall, published by John Wiley & Sons, 2011."

Motor 5
Shift frequency 454 Hz - 12100 Hz

FAULT

STATUS

Imbalance → WARNING

Misalignment → WARNING

Ready to Fly → READY

Imbalance

On the spectral difference plot we can see Motor 3's 1x shaft peak is significantly higher than the baseline peak. This passes the orange line.

Misalignment

On the spectral difference plot we can see Motor 3's 1x shaft peak is significantly higher than the baseline peak. This passes the orange line.

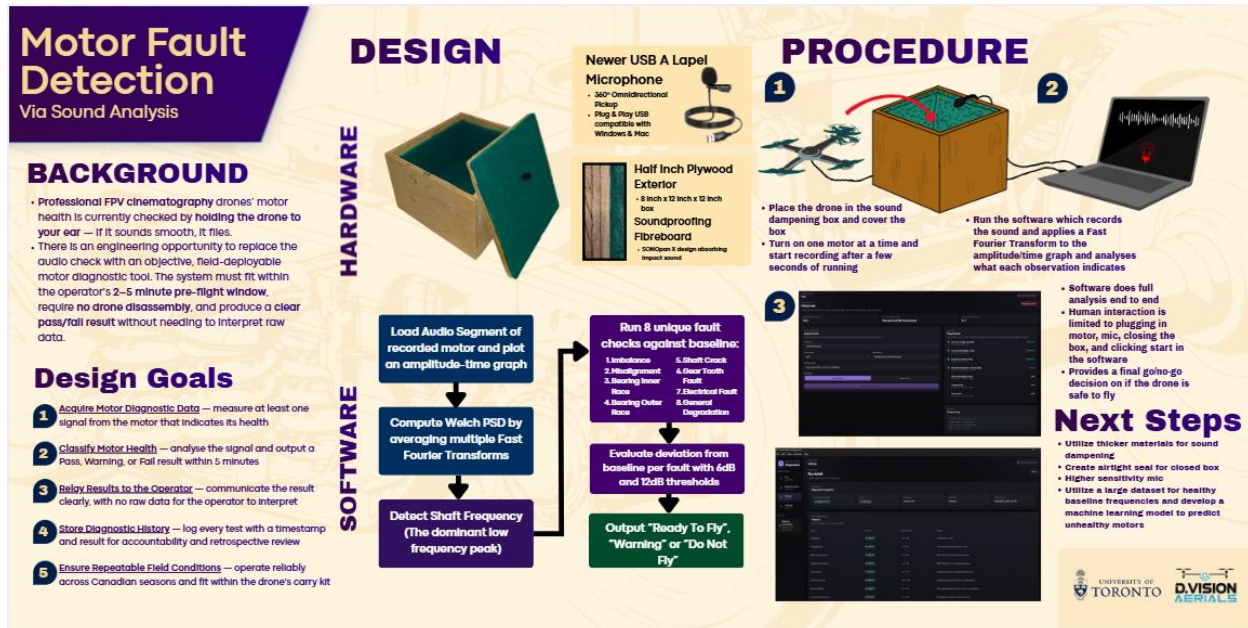
Verification

- Signal acquisition** — Recorded audio from each motor individually
- Fault detection and calibration** — Compared each motor against healthy baseline
- Sensitivity to faults** — Analyzed changes in FFT
- Response time** — Achieved ~3-4 seconds
- Non-invasive installation** — No physical modification (no soldering/drilling)
- Network independence** — No need for Internet access
- Output clarity** — Presented results to users as Ready to Fly / Warning / Do Not Fly

Supports:

This extract supports communication and representation claims by showing the concise stakeholder-facing summary of the converged concept.

Extract:



Supports: This extract supports synthesis claims by showing how project evidence and conclusions were condensed for presentation.

Extract: **5 Opportunity Needs, Goals, and Objectives**



Figure 1: Flowchart depicting needs, goals, and objectives for addressing the opportunity.

Supports: This extract supports framing and scope claims by showing the requirements/context that anchored the final diagnostics direction.

15 Praxis II Team — Verification Processes

Extract:

Before Going to MYFAB

- Change NUM_SAMPLES to 512 in both Arduino sketch and analyze.py
- Re-upload sketch to Arduino Mega
- Install dependencies: `pip install pyserial numpy matplotlib`

Physical Setup

- Green alligator clip (signal) → Arduino A0 — the very bottom pin of the analog header, right next to the power pins
- Black alligator clip (ground) → Arduino GND pin in the power section
- Arduino → laptop via USB
- Confirm CH1 output button on generator is illuminated

Generator Settings (all tests use CH1 BNC output)

Test	Frequency	Amplitude	Offset	Sum Mod	CH2
Healthy clean	1332 Hz	1.0 Vpp	+0.5V	Off	Off
Faulty clean	1332 Hz	1.0 Vpp	+0.5V	Sine, 3730 Hz, 1%	Off
Faulty + 5% noise	1332 Hz	1.0 Vpp	+0.5V	Sine, 3730 Hz, 1%	Noise 0.05Vpp, Combine ON
Faulty + 20% noise	1332 Hz	1.0 Vpp	+0.5V	Sine, 3730 Hz, 1%	Noise 0.20Vpp, Combine ON

Run Commands

```
python analyze.py --sensor HEALTHY_CLEAN --port COM6
python analyze.py --sensor FAULTY_CLEAN --port COM6
python analyze.py --sensor FAULTY_NOISY_5PCT --port COM6
python analyze.py --sensor FAULTY_NOISY_20PCT --port COM6
```

Each Test — Check These in the Plot

- Large spike sitting on the green 1332 Hz line ✓
- Fault tests: smaller spike on the red 3730 Hz line ✓
- Noise tests: raised noise floor but fault peak still visible ✓
- Console shows time under 15 seconds ✓

You Leave With

- 4 PNG plots
- 4 .npy raw data files (lets you re-run analysis at home)
- Console output showing SNR and timing verdicts for each test

Supports: This extract supports validation claims by showing the verification logic and testing approach used to assess prototype behavior and limitations